

HARDWARE SDK · GUIDE

Set up & use your WiFi photo frame

Turn a ~\$10–14 ESP32 board and a small TFT screen into a digital photo frame that receives photos over WiFi straight from the **ProFocus** app — with zero app changes.

ESP32 / Arduino

Receive photos over WiFi

5 hardware tiers

No cloud · LAN-only

1 How it works

The ProFocus app already shares photos to another device over your local WiFi using a tiny HTTP server on **port 8080**. This SDK makes a cheap microcontroller **look exactly like that server** — to the app, your DIY frame is just "another ProFocus screen" on the network. Send one photo → it appears on the frame instantly and is saved to the memory card.

Send from the app

Open photo sharing in ProFocus, pick the frame by its IP (or scan the QR on the frame's screen), send.

Straight over WiFi

JPEG over HTTP on your LAN. No internet, no account, no cable.

Slideshow + remote

The frame cycles photos and obeys next / prev / pause from the app. Album survives a reboot.

2 Pick your hardware (cheap → premium)

Five builds, ordered cheap/fewer-features → pricey/more-features. New here? Choose **Tier 2**.

Tier	Board + screen	Phone push?	Album	~Cost
0 · Starter (offline)	Arduino Uno + 3.5" TFT + SD	✗ SD copy only	SD card	\$8
1 · Essential WiFi	ESP32 + 2.4"/2.8" ILI9341	✓ WiFi	SPIFFS (~10)	\$10
2 · Recommended ★	ESP32 + 2.8"/3.5" ILI9341 + microSD	✓ WiFi	SD (60+)	\$14
3 · Round clock	ESP32 + 1.28"/2.1" GC9A01 round	✓ WiFi	SPIFFS/SD	\$13
4 · Premium (hi-res)	ESP32-S3 + 3.5"/4.0" ILI9488 + microSD	✓ WiFi	SD (100+)	\$22

Note: the Arduino Uno (Tier 0) has **no WiFi**, so the phone can't push to it directly — it's the offline learning build. For "tap send on your phone", pick an ESP32 (Tier 1–4).

Quick shopping list (Tier 2)

- **ESP32 DevKit WROOM-32** — search "ESP32 DevKit WROOM-32" on AliExpress / Amazon / local shops (~\$6–9)
- **2.8"/3.5" ILI9341 SPI screen** (prefer one **with a built-in microSD slot**) — search "2.8 ILI9341 SPI TFT SD" (~\$5–8)
- **microSD card 8–32GB** (FAT32), female-female jumper wires, USB 5V cable/power

Full bill of materials + buy links for every part type (board, screen, accessories, 3D-printed case) are in `profiles/README.md` in the SDK.

3 Wiring — Tier 2 (ESP32 + ILI9341)

TFT pin	ESP32	TFT pin	ESP32
VCC	3V3	SDI / MOSI	GPIO 23
GND	GND	SCK	GPIO 18
CS	GPIO 5	SDO / MISO	GPIO 19
RESET	GPIO 4	LED / BL	3V3 (or GPIO 32 for brightness)
DC / RS	GPIO 2	SD_CS	GPIO 15

Pins for the other builds (Tier 3 round, Tier 4 ESP32-S3) are in the matching JSON file in `profiles/`. Power the screen from **3V3**, not 5V.

4 Flash the firmware (ESP32, Tier 1–4)

- 1 **Install Arduino IDE 2.x** and the **ESP32 board package** (Boards Manager → type "esp32" by Espressif).
- 2 **Install 3 libraries** (Library Manager): `TFT_eSPI` (Bodmer), `TJpg_Decoder` (Bodmer), `QRCode` (ricmoo).
- 3 **Open** `firmware/esp32-frame/esp32-frame.ino`.
- 4 **Configure the display**: copy `User_Setup/profile-B-ili9341.h` (or C/D) over `TFT_eSPI/User_Setup.h`. *Most-forgotten step — wrong driver = white screen.*
- 5 **Edit `config.h`**: set `WIFI_SSID` / `WIFI_PASSWORD`, choose `FRAME_PROFILE` (and `USE_SD 0` for Tier 1 without a card).
- 6 **Select the board** (e.g. "ESP32 Dev Module"), pick the COM port, click **Upload**. The frame joins WiFi and shows its **IP + a QR code** on the screen.

Offline Arduino Uno build (Tier 0): see `firmware/arduino-uno-sd/` — copy 24-bit BMP images to the SD card from a computer.

5 Send photos from the ProFocus app

- 1 **Power the frame** and wait for it to show `IP:8080` and a QR code (e.g. `192.168.1.42:8080`).
- 2 **Open the ProFocus app** on your phone → go to **photo sharing / Local Sync**.
- 3 **Pick the receiver**: open "Recent Devices", **enter** the frame's `IP:8080` (or use the camera to **scan the QR** on the frame).
- 4 **Select photos** and tap **Send**. They appear on the frame immediately and are saved to the card.
- 5 **Remote (optional)**: use the app's remote screen to send **Next / Prev / Pause**.

No app? Open `http://IP:8080` in any phone browser and upload there. Or use the `sender/` page bundled in the SDK (auto-resizes before sending + has remote buttons).

6 Protocol — integrate into your own app

If you write your own sender app, it only needs to speak these few HTTP calls (the same contract the ProFocus app already uses):

```
POST /upload          # phone → frame: one photo (raw JPEG bytes)
  Content-Type: image/jpeg
  X-Filename: vacation.jpg    → 200 OK "OK", frame displays it

POST /cmd {"cmd":"frame.next"} # remote: next / prev / pause
GET  /state      # {"screen":"frame","running":true,...}
GET  /           # browser upload page (any phone)
```

Port 8080 is fixed. Discovery is by IP or QR (the frame prints both on screen) — no cloud, no mDNS required. Full details: `protocol/PROTOCOL.md`.

7 Troubleshooting

Symptom	Cause & fix
White / noisy screen	Wrong <code>User_Setup.h</code> (step 4) or wrong power. Check the correct driver + wire to 3V3.
Can't join WiFi	ESP32 is 2.4GHz only . Or set <code>FALLBACK_SOFTAP 1</code> , connect the phone to AP <code>ProFocus-Frame</code> (password <code>12345678</code>), send to <code>http://192.168.4.1:8080</code> .
App says OK but nothing shows	Image too large or not a baseline JPEG. Use the <code>sender/</code> page (auto-resize) or send JPEG $\leq$ screen width. Firmware still returns OK so the app doesn't hang.
Clock overlay wrong time	Uncomment the <code>setenv("TZ", ...)</code> line in <code>setup()</code> for your timezone (VN = <code>ICT-7</code>).
No SD card	Set <code>USE_SD 0</code> in <code>config.h</code> to use SPIFFS (fewer photos).
Phone can't find the frame	Phone and frame must be on the same WiFi . Enter the exact IP shown on screen (it can change after a router restart).

ProFocus Frame SDK — MIT License © 2026 Fighttech. Source, firmware, hardware profiles and the latest release at github.com/fighttechvn/profocus · Online guide & blog: profocus.fighttech.vn/sdk/. Get the ProFocus app on the App Store & Google Play.